

Assignment No. 6

A **manifest file** is a JSON file (manifest.json) that provides metadata about a web application. It is essential when developing Progressive Web Apps (PWAs) as it allows the app to behave like a native app on mobile and desktop platforms. The manifest file improves the app's responsiveness and installability.

Key Benefits of a Manifest File:

1. **Installability:** Enables the "Add to Home Screen" functionality for PWAs, allowing users to install the app like a native app.
2. **App Metadata:** Provides details like the app's name, icons, theme color, and start URL.
3. **Improved User Experience:** Defines the display mode (e.g., fullscreen, standalone) to enhance the app's behavior on various devices.
4. **Offline Access:** Works in conjunction with a service worker to provide offline functionality.
5. **Customizable App Look:** Specifies custom icons and splash screens for different devices and resolutions.

Steps to Add a Manifest File to a Responsive Web App:

1. Create the Manifest File:

- Create a file named manifest.json in the root of your project or a suitable folder (e.g., /public or /static).
- Add the following key properties:

```
{  
  "short_name": "MyApp",  
  "name": "My Responsive Web App",  
  "icons": [  
    {
```

```
"src": "icons/icon-192x192.png",
"type": "image/png",
"sizes": "192x192"
},
{
  "src": "icons/icon-512x512.png",
  "type": "image/png",
  "sizes": "512x512"
}
],
"start_url": "/index.html",
"display": "standalone",
"background_color": "#ffffff",
"theme_color": "#000000"
}
```

2. Include the Manifest File in the HTML:

- Link the manifest file in the <head> section of your index.html or main HTML file:

```
<head>
<link rel="manifest" href="/manifest.json">
<meta name="theme-color" content="#000000">
</head>
```

3. Add Icons:

- Provide icons in various resolutions for the app (e.g., 192x192 and 512x512 pixels). Place them in the specified folder (/icons/ in this example).
- Ensure the paths in manifest.json match the actual locations of the icons.

4. Configure a Service Worker (Optional but Recommended):

- Set up a service worker (sw.js) to handle caching and offline support.
- Register the service worker in your app's JavaScript file:

javascript

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```
if ('serviceWorker' in navigator) {  
  navigator.serviceWorker.register('/sw.js').then(() => {  
    console.log('Service Worker Registered');  
  });  
}
```

5. Test the Manifest File:

- Use Chrome DevTools to test the manifest file:
 1. Open your web app in Google Chrome.
 2. Go to **DevTools** → **Application** → **Manifest**.
 3. Verify that all properties, icons, and links are correctly configured.

6. Verify Responsiveness:

- Test the app on various devices and screen sizes to ensure that it responds well and meets PWA standards.

manifest.json:

```
{  
  "name": "React Web Applicaton For Car Rental Service",  
  "short_name": "CAR Rental",  
  "icons": [  
    {  
      "src": "/icon144.png",  
      "type": "image/png",
```

```
    "sizes":"144x144"
  },
  {
    "src":"/icon192.png",
    "type":"image/png",
    "sizes":"192x192"
  }
],
"screenshots":[
  {
    "src":"/screenshot1.png",
    "type":"image/png",
    "sizes":"320x320",
    "form_factor":"wide"
  },
  {
    "src":"/screenshot1.png",
    "type":"image/png",
    "sizes":"320x320",
    "form_factor":"narrow"
  }
],
"start_url":"/index.html",
"display":"fullscreen",
"orientation":"portrait"
}
```

Output:

The image displays a web browser window with a car rental advertisement on the left and the Chrome DevTools Application panel on the right.

Advertisement:

- CAR Rental** logo.
- Text: "Plan your trip now"
- Large text: "Save big with our car rental"
- Text: "Rent the car of your dreams. Unbeatable prices, unlimited miles, flexible pick-up options and much more."
- Buttons: "Book Ride" (orange) and "Learn More" (black).

Chrome DevTools Application Panel:

- App Manifest:** [manifest.json](#)
- Identity:**
 - Name: React Web Application For Car Rental Service
 - Short name: CAR Rental
 - Description:
 - Computed App ID: <http://localhost:3000/index.html> [Learn more](#)
 - Note: id is not specified in the manifest, start_url is used instead. To specify an App ID that matches the current identity, set the id field to `/index.html`.
- Presentation:**
 - Start URL: </index.html>
 - Theme color
 - Background color
 - Orientation: portrait
 - Display: fullscreen
- Protocol Handlers:**
 - Define protocol handlers in the [manifest](#) to register your app as a handler for custom protocols when your app is installed.
 - Need help? Read [URL protocol handler registration for PWAs](#).
- Icons:**
 - ☐ Show only the minimum safe area for maskable icons
 - Need help? Read the [documentation on maskable icons](#).
 - 144x144px image/png:
 - 192x192px image/png:
- Window Controls Overlay:**

Application

Manifest

Service workers

Storage

Storage

Local storage

Session storage

Extension storage

IndexedDB

Cookies

Private state tokens

Interest groups

Shared storage

Cache storage

Storage buckets

Background services

Back/forward cache

Background fetch

Background sync

Bounce tracking miti...

Notifications

Payment handler

Periodic background ...

Speculative loads

Push messaging

Reporting API

Need help? Read [Customize the window controls overlay of your PWA's title bar.](#)

Screenshot #1

Form factor wide

320x320px

image/png



Screenshot #2

Form factor narrow

320x320px

image/png

